

Yunmu Shu

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EDUCATION

Northeastern University

Master of Science in Computer Science (GPA: 3.9/4.0)

Boston, MA

Sep. 2022 – Aug. 2025

Northeastern University

Master of Science in Financial Mathematics

Boston, MA

Sep. 2021 – Aug. 2022

Guangdong University of Foreign Studies

Bachelor of Science in Economics

Guangzhou, Guangdong

Sep. 2017 – Jun. 2021

EXPERIENCE

Software Developer Intern

Jan. 2025 – Jun. 2025

First Help Financial

Needham, MA

- Engineered backend modules for an auto loan origination and decisioning platform used by dealers and underwriters, enhancing automation and ensuring regulatory compliance throughout the funding evaluation workflow
- Analyzed slow SQL queries in loan transaction module using MySQL EXPLAIN, added composite indexes and optimized pagination with keyset query logic, reducing query latency by **22%** under 10k+ records
- Built a centralized Vehicle Data Retrieval service integrating Elasticsearch and **Redis** caching, boosting query throughput by 15% and handling 500k+ monthly search requests across multiple business modules
- Monitored and visualized database performance metrics via Prometheus and Grafana to reduce overall CPU utilization

Software Developer Intern

Feb. 2024 – Jun. 2024

Dassault Systèmes

Waltham, MA

- Contributed to backend development of ENOVIA document workflow platform within the 3DEXPERIENCE ecosystem, supporting regulatory compliance and enterprise collaboration
- Architected and deployed **role-based access control** and audit logging for secure document approval and traceability
- Built and optimized an **API Gateway** for multi-system document synchronization and metadata retrieval, integrating authentication and routing rules to improve throughput by **28%**
- Introduced centralized **caching layer** and API rate limiting to enhance system scalability and prevent redundant access
- Developed asynchronous job handlers using Java Servlet and MQL to process large-scale document import tasks

Software Developer Intern

Jun. 2024 – Aug. 2024

Rapid Micro Biosystemes

Lexington, MA

- Enhanced backend architecture of a laboratory imaging platform for automated petri dish image capture and data storage
- Optimized schema by restructuring entity mappings and introducing partitioning strategy to accelerate data retrieval by 25%
- Refactored legacy PHP codebase into modular service components and integrated automated data migration scripts in Bash, ensuring consistency across **1M+** image records and improving maintainability and release stability
- Containerized development environment with **Docker** and Apache to replicate production configuration

Software Developer Intern

May. 2023 – Aug. 2023

Tatfook Technology Co., Ltd

- Developed a responsive reporting dashboard for user-generated content analytics using **Next.js**, enabling real-time data visualization, trend tracking, and automated report generation for application users
- Refined server-side rendering and API data fetching with **Axios**, reducing average page load time by **30%**
- Configured CI/CD pipelines utilizing **GitLab** for code uploads and **Jenkins** for seamless deployment

PROJECTS

Tripadvisor App

Oct. 2024 – Nov. 2024

- Developed a Tripadvisor-like web application using **Redis** as middleware to handle high-concurrency scenarios
- Engineered flash-sale module with read/write separation and atomic transaction logic through Redis **Lua scripting**
- Integrated distributed **Kafka** message queue for asynchronous order processing, reducing API latency to tens of milliseconds
- Designed user engagement features such as like/follow using **ZSet** and check-in tracking via **Bitmap** for efficient query

GeeCache

Jun. 2024 – Jul. 2024

- Implemented a distributed caching system in Go, developing both local in-memory cache and HTTP-based cache with **Least Recently Used** strategy, and improved inter-node communication using **protobuf**
- Employed **Mutex** locks to prevent cache breakdown and ensure data consistency under concurrent access
- Integrated consistent hashing to select nodes, ensuring load balancing and efficient cache distribution

TECHNICAL SKILLS

Programming Languages:

Frameworks:

Databases & Caching:

Cloud & Tools:

Java, C/C++, Go, TypeScript, JavaScript, SQL, Python
Spring Boot, Spring Cloud, React.js, Next.js, Express.js, Node.js
MySQL, Redis, MongoDB, PostgreSQL, DynamoDB, Kafka
AWS, Docker, Kubernetes, Jenkins, Nginx, Git, Terraform